

Imaging exoplanets beyond the telescope stability limit with differentiable physical models

Abstract

The direct imaging of exoplanets is the primary method for detailed characterization of planets orbiting beyond 1 au¹, but it requires suppressing the bright glare of the star to see the faint planet light. Sophisticated instrumentation and image processing are both necessary for effective starlight suppression. While advances in coronagraphy² and telescope stability³ are increasing the sensitivity of our observations, image processing continues to be a significant limitation. This is because standard algorithms use reference images for starlight modeling, and thus they are limited by inevitable telescope instabilities that change from reference to science observation⁴. As a result, standard starlight removal is almost an order of magnitude away from the fundamental noise floor⁵, limiting the sensitivity of current telescopes. Addressing this challenge will increase the number of planets amenable to direct imaging, and will be a major step towards imaging exo-Earths with future observatories⁶. Here we present a new physics-based, GPU-accelerated method for starlight subtraction. Our method explicitly models the physics of the optical system, including telescope instabilities, to produce an accurate starlight model. The key innovation that makes this feasible is a differentiable optical model enabled by modern machine learning frameworks with automatic differentiation⁷, allowing efficient optimization over millions of degrees of freedom. We demonstrate our method on observations from the James Webb Space Telescope (JWST). We recover previously undetected planets and improve detections of known planets with sensitivity 2-10 times superior to standard

techniques. Our method immediately increases the exoplanet discovery potential of JWST and highlights how machine-learning-enabled optimization can pave the way for characterization and compensation of telescope instabilities. Our framework can be extended to ground-based observatories incorporating atmospheric turbulence measurements, and to future space-based observatories like the Roman Space Telescope and the Habitable Worlds Observatory, potentially relaxing the unprecedented stability currently required to make these a success.

1 Main

While exoplanet imaging has greatly advanced our understanding of planetary astrophysics (e.g., discovering planets at wide orbital separations^{8,9}, characterizing atmospheric physics with emission spectroscopy^{10–13} and studying still-forming protoplanets^{14,15}), the number of imaged planets remains small. An important reason for this is the challenging contrast between the brightness of planets and their host stars: current instruments can only image the brightest young giant planets, and even they are 10^4 to 10^6 times fainter than their stars in the near infrared.

In order to detect any planetary signals, the glare of the star (or stellar point spread function, PSF) must be modeled and removed from our observations. Modeling the stellar PSF to the required precision is difficult due to its strong dependence on stellar properties and instrumental aberrations. Until now, the most successful approaches build the PSF model empirically using a library of reference observations taken at other times or wavelengths, or of other stars (e.g., ^{16,17}). These empirical methods have been the standard in exoplanet imaging for decades, and together with advances in instrumentation and observing strategies they have enabled the discovery and study of the brightest wide-separation giant planets with direct imaging.

49 However, these empirical approaches are fundamentally limited by instrument
50 instabilities, such as telescope pointing errors and thermal drifts. These instabilities
51 cause changes in the PSF between reference observations and the science observation
52 of interest (Figure 1), reducing both the accuracy of the constructed PSF model and
53 the detection sensitivity of our observations^{18,19}. For this reason, one of the main
54 challenges for future missions like the Habitable Worlds Observatory (HWO), with
55 the ambitious goal of imaging Earth analogs around Sun-like stars (10^{10} brightness
56 contrast), is achieving an extreme and unprecedented stability at the picometer level
57 (100 to 1000 times more stable than JWST^{4,20}). Advances in PSF modeling that do
58 not rely as heavily on observatory stability could have transformative implications for
59 missions like HWO, potentially relaxing the stringent stability requirements needed
60 to image exo-Earths. Even for current facilities, like JWST or ground-based observa-
61 tories, overcoming instrumental instabilities could drastically increase the number of
62 planets amenable to direct imaging.

63 An attractive alternative approach is model-based PSF subtraction: directly mod-
64 eling the physics of the instrument to reconstruct the stellar PSF. This approach has
65 been noted for its potential to relax the stability requirements of the telescope, to
66 reduce required exposure times and for the possibility of reaching the fundamental
67 performance limit set by Poisson noise⁶. While promising, model-based PSF subtrac-
68 tion is a formidable challenge given the complexity of optical systems that prevents
69 accurate modeling^{21,22}. Previous work investigating this approach on ground-based
70 telescopes highlighted exceptional computational difficulties²³.

71 Fortunately, recent years have seen extraordinary progress by the machine learning
72 community in imaging problems. Crucial to the success of these efforts, from optical
73 microscopy to computer graphics, has been GPU-accelerated differentiable models.
74 Differentiable models are models that leverage automatic differentiation⁷, or autodiff,
75 to compute analytic derivatives even with millions of free parameters. Autodiff has

76 been the backbone behind the successful training of modern large neural networks, in
77 addition to enabling very efficient optimization for physical models. Autodiff appli-
78 cations in astronomy have started to be explored only recently, with some examples
79 including wavefront control²⁴ and speeding up modeling of exoplanet transits²⁵, grav-
80 itational lensing systems²⁶ and stellar population synthesis²⁷. Autodiff has also been
81 recently noted for its ability to measure and fix systematic errors in non-coronagraphic
82 modes of JWST²⁸, with clear implications for instrument calibration.

83 This paper reports the successful application of automatic differentiation in coron-
84 agraphic exoplanet imaging, making model-based PSF subtraction feasible for the first
85 time: Physics-based Learning for Imaging Planets (FLIP). By simulating the optical
86 system, FLIP solves for telescope aberrations (e.g., instantaneous pointing and phase
87 errors), thus overcoming instrument instabilities that limit traditional PSF modeling
88 techniques. By using automatic differentiation, FLIP overcomes previous computa-
89 tional challenges to achieve the necessary PSF model fidelity. We demonstrate the
90 effectiveness of FLIP by using it to directly image exoplanets in real data from the
91 JWST Near Infrared Camera (NIRCam) coronagraph instrument. FLIP produces bet-
92 ter planet detections than standard methods in published datasets, and in some cases
93 it reveals images of planets that had remained undetected by traditional techniques.

94 After over thirty five years of repeated community efforts²⁹, this work is the first
95 time that exoplanets are revealed in coronagraphic imaging data using a physical model
96 instead of empirical reference observations. This is also the first time that model-based
97 PSF subtraction surpasses standard techniques, with up to an order of magnitude
98 improvement in sensitivity. These results show the potential of physics-based, autodiff-
99 enabled algorithms to surpass limits set by telescope stability in current and future
100 observatories.

1.1 FLIP method overview

FLIP, Physics-based Learning for Imaging Planets (Figure 1), consists of two primary innovations: *physics-based modeling*, to directly model instrument instabilities, and *automatic differentiation* to address the computational challenge of achieving the necessary PSF model accuracy.

Building on previous work on simulated monochromatic data³⁰, we build a realistic differentiable model to capture the complexities of real JWST data. Our model consists of a physical scene (a point source corresponding to the host star), an optical model and a detector model. Our JWST optical models are based on the nominal models provided in the open-source software STPSF³¹, the *de facto* JWST PSF modeling tool maintained by the Space Telescope Science Institute. These models are known to be imperfect, particularly for NIRCcam coronagraphy, but they represent the current best characterization available of the optics and known aberrations. With this model as our initial condition (implemented in PyTorch³²), we solve for wavefront aberrations, the position and flux of the star and the pixel-level response of the detector. See Methods for details.

We implement an optimization framework to maximize sensitivity and minimize overfitting consisting of two stages: full optimization on a reference star (typically a bright star taken close in time to the science target, likely with no planets or binary stars around it), followed by further tuning on the science target with a lower learning rate. We assess a stopping criterion by injecting synthetic planets close to the sensitivity limit and tracking the signal-to-noise ratio (SNR) of the fake planet detection. We halt the optimization when its SNR reaches a maximum. This also allows us to quantify the extent of overfitting in the final PSF-subtracted image. This end-to-end optimization, tuning around 10^6 free model parameters, takes under 35 minutes with FLIP in a regular NVIDIA A30 GPU. See Methods for details.

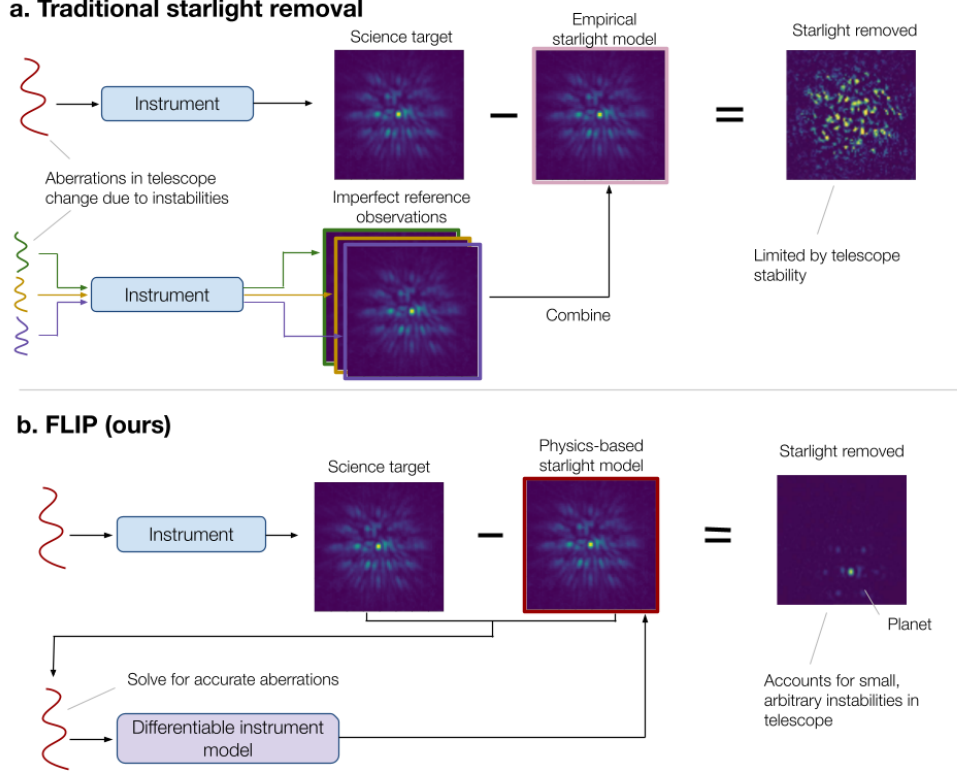


Fig. 1 Schematic comparison of standard starlight subtraction and our method based on physical modeling, FLIP. Standard methods are limited by inevitable instabilities in the telescope, thus producing imperfect PSF subtraction. FLIP allows solving for those small changes in the telescope aberrations, circumventing the stability limit of the instrument.

1.2 Results

We show the performance of FLIP on published JWST NIRCcam data, and we compare it with the optimal PSF subtraction obtained in those publications with standard method KLIP¹⁷. We analyze observations of two known planets: HIP 65426 b³³, first published in³⁴, and 51 Eridani b³⁵ in two filters originally published in³⁶. We additionally process an observation of TYC 5899, a target from planet search survey with no known companions³⁷, to showcase the exoplanet discovery potential of FLIP. Figure 2 shows comparisons of the PSF-subtracted observations using KLIP and our method FLIP.

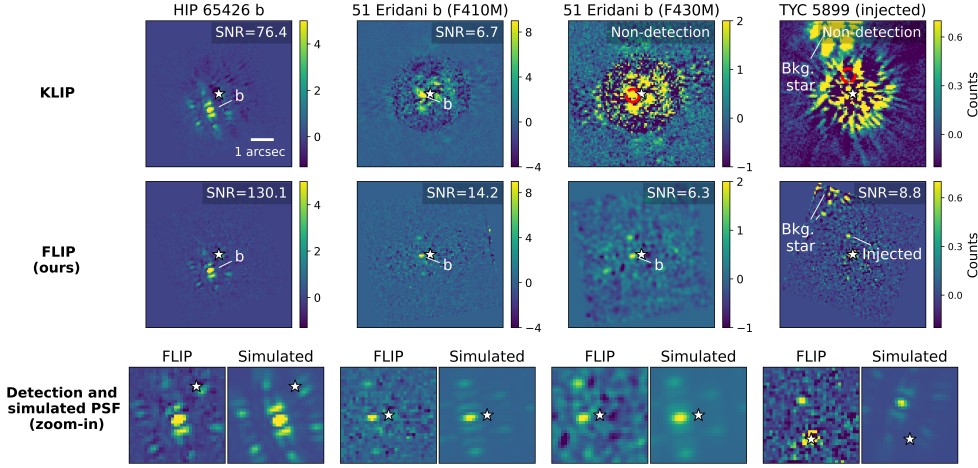


Fig. 2 Our method, FLIP, compared to the state of the art method KLIP. Top: Data processed with KLIP, consistent with published results. Middle: Data processed with FLIP, showing the full field of view optimized over (80×80 pixels). Bottom: Our detected companions with FLIP next to ideal simulated PSFs, to highlight the recovered morphology and to distinguish real point sources from PSF side lobes. The position of the star is marked with a star-shaped icon. In all panels, North is up and East is left. 51 Eridani F430M observations are convolved with a Gaussian kernel ($\sigma = 1$ px) to smooth out pixel-to-pixel noise (only in this panel is pixel-to-pixel noise comparable with the brightness of the companion). All panels are in units of flux density (MJy/sr except in 51 Eri F430M, which had MJy/sr units before smoothing).

FLIP achieves superior performance in all observations relative to KLIP, with large improvements (4-10x) in some cases. We obtain around 1.7 times better SNR on HIP 65426 b, 2.1 times better SNR on 51 Eridani b in F410M, and a robust detection of 51 Eridani b in F430M, which yielded a non-detection in the original publication (sensitivity was improved by a factor of ~ 4 at the location of the planet). For TYC 5899, we inject fake planets on both data sets to illustrate the sensitivity improvement. One of the injections is shown in the figure, which we recover at high significance while it is far from being detected with KLIP. Additionally, there is a bright, nearby star in the TYC 5899 scene which is still visible after our PSF subtraction, but it has been distorted. See Methods for details.

Our new detection of 51 Eridani b in F430M is the first image of the planet at $4.3 \mu\text{m}$, where the absorption band of carbon dioxide is located. This is a wavelength range inaccessible to ground-based measurements due to telluric contamination from the

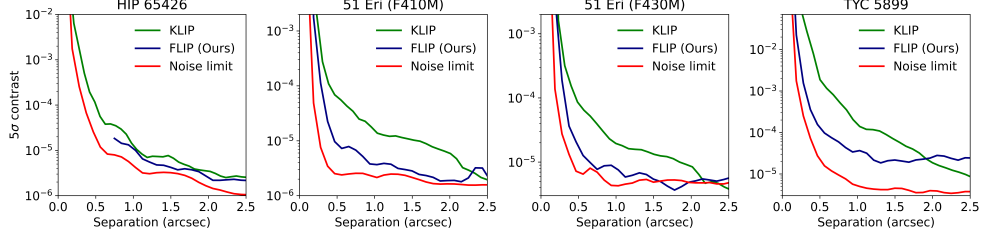


Fig. 3 Detection sensitivity curves for KLIP and FLIP, calibrated with injection-recovery tests. Also shown is the fundamental noise-limited sensitivity. Our method enables significant improvements (in some cases 4-10x) in exoplanet detection sensitivity.

Earth’s atmosphere. With CO_2 being a good tracer of atmospheric metallicity, future work will use this detection to measure the $4.3 \mu\text{m}$ photometry of the planet and carry out a new atmospheric characterization. This will help assess previous evidence for an elevated metallicity of about 5 times the stellar value^{12,36} and better understand the formation history of 51 Eridani b.

Figure 3 shows significant improvement in sensitivity to exoplanet detection compared to KLIP. These results are illustrative of the potential of FLIP to aid exoplanet discovery with JWST. Several current exoplanet discovery surveys follow similar observing strategies to the TYC 5899 dataset in particular (e.g., JWST programs GO 5835, SURVEY 6005), with sub-optimal reference observations to increase observing efficiency. Our results show that FLIP is especially beneficial in these cases, with sensitivity gains of up to an order of magnitude. With more than 100 observations taken in these programs, the exoplanet discovery potential of using FLIP in these surveys is large.

1.3 Future improvements

By being insensitive to instrument instabilities in principle, only model-based PSF subtraction is capable of reaching sensitivities down to the fundamental noise limit⁶. FLIP, in general, achieves contrast levels much closer to the noise limit than standard method KLIP (Figure 3), but the photon noise is not reached in any observation

168 yet. Clear avenues for improvement include a more sophisticated detector model to
 169 process deep observations of very bright stars^{28,38}, disentangling degeneracies between
 170 different model parameters, reducing the computational cost of the method (currently
 171 the runtime of FLIP on an NVIDIA A30 GPU is 15 to 30 minutes) and a framework
 172 to handle large data volumes for ground-based applications. See Methods for details.

173 1.4 Discussion

174 By explicitly modeling the physics of the instrument and leveraging optimization tools
 175 from the machine learning community, our technique FLIP achieves exoplanet detec-
 176 tion sensitivity beyond the stability limit of JWST. This has implications in the short
 177 term and in the long term for high-contrast imaging.

178 In the short term, the 2–10× improvement in sensitivity shown in Figures 2 and
 179 3 is significant for exoplanet discovery and characterization with JWST. With more
 180 than 100 stars being observed in planet search surveys with JWST, FLIP can improve
 181 the discovery sensitivity in nearly all of these observations, uncovering fainter planets
 182 (that is, colder and lower mass) that were previously inaccessible (Figure 3) and ruling
 183 out false positives from starlight residuals that may look like planets. By accessing
 184 even lower temperature and lower mass planets (precisely the parameter space that
 185 JWST is uniquely sensitive at), FLIP will contribute to maximizing the exoplanet
 186 legacy of JWST. Furthermore, the upcoming Data Release 4 of the Gaia mission is
 187 predicted to discover thousands of giant planets^{39,40}, the vast majority of which will
 188 be at separations < 5 au. These are very close separations even for very nearby stars
 189 (< 0.5 arcsec at 10 pc), precisely where FLIP shows the best gain in performance
 190 (Figure 2). FLIP could therefore be instrumental to unlocking the true sensitivity of
 191 JWST to characterize the atmospheres of the Gaia planets.

192 In the long term, these first results show the potential of physics-based image
 193 processing on other current and future telescopes, as well as the crucial role that

194 machine-learning-based techniques can play. For ground-based observatories, our
195 method can be coupled with adaptive optics systems and real-time wavefront sensing
196 data to solve for internal unknown instrument drifts. While this involves distinct com-
197 putational and data volume challenges (see Methods), this extension of our work could
198 greatly increase the sensitivity of ground-based data, particularly if adaptive optics
199 telemetry is stored and made publicly available. The unique potential of model-based
200 PSF subtraction to get to the photon noise limit could be instrumental to directly
201 image temperate terrestrial planets around low mass stars with the extremely large
202 telescopes^{41,42}.

203 For space-based missions, the extreme levels of stability needed to image exo-Earths
204 with the future Habitable Worlds Observatory are a major challenge^{43,44}, and one that
205 aggressively restricts the design space for the telescope. In this work we have shown for
206 JWST, however, that model-based PSF subtraction can model these very instabilities
207 and achieve contrasts better than the limit set by the thermal and mechanical drifts
208 of the telescope. If this approach can be extended to HWO contrast levels, the ultra-
209 stability requirement for the system may be partially lifted and make the design of the
210 telescope more tractable. Investigating the applicability of this technique to HWO will
211 be critical, in particular through testing in the upcoming Roman Space Telescope⁴⁵
212 and the Lazuli Space Telescope⁴⁶. If successful, the goal of imaging true exo-Earths
213 and achieve photon noise-limited high contrast imaging for the first time will be closer
214 to reality.

215 **2 Methods**

216 **2.1 The FLIP framework**

217 **2.1.1 Model design**

218 FLIP models the stellar PSF by optimizing many of the free parameters in a physical
219 model that influence the PSF, including the star being observed and the instrument
220 used for the observation. We base our models on the nominal **STPSF** optical models,
221 which are used to simulate PSFs in JWST for a variety of instruments and filters.
222 Extended Data Figure 4 shows a schematic diagram equivalent to the physical models
223 used to simulate observations with the JWST NIRCam coronagraphs.

224 Given these models, we find that four groups of free parameters are generally
225 sufficient for PSF fidelity in most cases:

- 226 • The flux of the star and its precise position behind the focal plane mask of the
227 coronagraph.
- 228 • The optical path difference, or OPD (a phase aberration) before the focal plane
229 mask of the coronagraph.
- 230 • The OPD after the focal plane mask of the coronagraph.
- 231 • The response per pixel in the detector.

232 Physically, these parameters capture the imperfect alignment of the star with the
233 coronagraph (telescope pointing errors), the imperfect alignment of the primary mir-
234 rors in JWST (deformations or tilts relative to the closest-in-time OPD measurement),
235 the imperfectly-known OPD internal to specific instruments in JWST (in our case,
236 the coronagraphs in the NIRCam instrument) and imperfectly-known detector cali-
237 brations. While other physical mechanisms are capable of affecting the PSF (such as
238 pupil shear, manufacturing errors in the optics or more complicated detector physics),

we find that our choice of free parameters is generally capable of effectively reproducing the effects of these other systematics and produce high quality PSF subtractions. A summary of our model parameters is shown in Extended Data Table 1.

Implementation details. Following STPSF, our model implements light propagation in the Fraunhofer approximation via Fourier optics. When modeling any specific observation, STPSF calculates the nominal instrument state to produce a simulated PSF, taking into account the instrument and filter, the OPD measurement closest to the time of observation and the spectrum of the host star to model both wavelength-dependent OPDs and the relative intensity contribution of each wavelength in broadband observations. We take this instrument state and we replicate the optical model in PyTorch³². See³¹ for details on the STPSF model construction.

On top of the nominal STPSF model, we perform a sub-pixel shift to align the center of the PSF (by default at the center of the output array in) to the center of the coronagraph on the detector as measured for a given coronagraph mask and filter. We do this to perform the optimization directly on the detector coordinates, and avoid artifacts induced by interpolating and shifting the data itself. Also following STPSF, the PSF is convolved with a small Gaussian kernel ($\sigma = 0.018$ arcseconds) to mimic a basic charge diffusion effect. Finally, each pixel is multiplied by a pixel response function.

We parametrize the OPDs as square arrays of 1024×1024 pixels, each pixel containing the OPD value in meters. Different parametrizations were tested (e.g., Zernike polynomials across the telescope aperture, Zernike polynomials on each segment of the primary mirror), but this formulation showed the best results.

Following recommendations from STPSF, we perform our simulations with pixel over-sampling of 2 in the image plane, and we use 21 monochromatic wavefronts for F444W observations and 9 for F410M and F430M for the best balance between numerical accuracy and evaluation speed.

2.1.2 Optimization design

Previous work on simulated monochromatic data³⁰ found that given a near perfectly accurate physical model, it is possible to optimize it directly on science observations and produce excellent PSF subtraction without overfitting planet signals. However, in our experiments with real data we find that accidentally subtracting planet signals is indeed possible. This is likely due to STPSF models having substantial inaccuracies relative to real JWST instruments, thus making the optimization problem less straightforward.

To address this, we implement a two-stage optimization. In stage 1, we leverage the full flexibility of our model and we calibrate it on a reference star, optimizing all the free parameters described in Section 2.1.1. This helps overcome inaccuracies in the optical model and sets the parameters closer to a good solution. In stage 2, we use our calibrated model and optimize only a reduced number of free parameters directly on the science observation. This accounts for different pointing errors and different OPDs between reference and science observations due to telescope instabilities.

In both stages, we optimize different parameters in separate sub-stages. This is done to reduce degeneracies between different parameters. In stage 1 we have three sub-stages: 1) solving for star position and flux, 2) solving for both OPDs, and 3) solving for both OPDs and pixel responses. In stage 2 we have two sub-stages: 1) solving for star position and flux, and 2) solving for the OPD after the focal plane mask. While the OPD before the coronagraph mask is also expected to change due to the JWST mirrors, we currently find the optimization to be more stable if we solve for the OPD after the mask instead. These stages are summarized and illustrated in Extended Data Figure 5.

Implementation details. We use the Adam optimizer with a Cosine Annealing learning rate scheduler. We use an L1 loss function for our optimization.

292 We tune the learning rates in our model by hand, increasing it across orders of
293 magnitude until we find one that produces a stable optimization with convergence
294 below around 15000 iterations. Due to the different magnitudes of different parameters
295 in our model, we assign a different learning rate to each family of parameters. A simple
296 coarse search clearly shows the appropriate scale for each parameter. We find that a
297 learning rate of 10^{-9} for both OPDs, 10^{-8} for the star position and 10^{-5} for detector
298 pixel responses show good performance in all our experiments. Our results are not
299 strongly sensitive to order unity changes in learning rates. As an illustration, Extended
300 Data Figure 6 shows order unity variations to learning rates do not significantly affect
301 our results, and that increasing or decreasing them by orders of magnitude clearly
302 shows the correct scale for stability and relatively fast convergence. A summary of the
303 order of magnitude of these learning rates per parameter is shown in Extended Data
304 Table 1.

305 Across different observations, we find that globally scaling all learning rates by
306 some order-unity constant can help with stability and convergence speed. This is likely
307 due to the magnitude of the loss function depending on the brightness of the stars.
308 We find that a global scaling of 4 works well in all the experiments presented here, but
309 preliminary analysis on fainter stars showed better results by using 1 or 0.5 instead in
310 those cases.

311 Similarly, during the last sub-stage of our optimization (fitting the OPD drift on
312 the science target), we reduce the global scaling factor on the learning rate to around
313 1% of its original value to further reduce risk of overfitting. Our results are not strongly
314 sensitive to the exact choice of this value, but it can be tuned differently for different
315 observations depending on the similarity between reference star and science target.

316 For calibration on the reference star (stage 1), we generally find that 600-800
317 iterations to fit stellar position and flux, 3500 to fit OPDs, and 1500 to fit OPDs and
318 detector responses produce good results across different observations. This process

319 takes around 10 to 15 minutes on a single NVIDIA A30 GPU. Once the model is
320 calibrated for some observation, the calibrated model can be saved for multiple runs.

321 For parameter tuning on the science target (stage 2), we find that 600-800 iterations
322 to fit stellar position and flux, and 5000 to 9000 iterations for OPD drift (depending
323 on convergence) show good results overall. This process takes 10 to 30 minutes on a
324 single NVIDIA A30 GPU.

325 To determine a stopping point for our optimization, we inject fake planets in the
326 science observation and we track their SNR as a function of iteration number. As
327 expected, the SNR of the injected planets increases for a number of iterations and
328 then decreases due to algorithm overfitting. Therefore we stop our optimization at the
329 approximate iteration when the injected planets reach a peak SNR, which was found
330 to be consistent for planets injected at similar separations.

331 While our optimization can be done on each individual frame of the reference or sci-
332 ence observations, we found that optimizing on the time-averaged observations yielded
333 better contrast results for much shorter computation time. This is probably because
334 KLIP subtractions are currently limited by static or very slowly-evolving aberrations
335 in JWST, as opposed to other shorter-timescale aberrations⁴. Time-averaged obser-
336 vations also have higher SNR on the static components of the PSF, likely leading to
337 better optimization results.

338 2.2 Experiments

339 We carried out experiments on four datasets: three with known companions (HIP
340 65426 in one filter and 51 Eridani in two filters) and one with no companions, part of
341 a planet search survey (TYC 5899). We run standard PSF subtraction method KLIP
342 and our technique FLIP on all four datasets and compare the performance of each,
343 both with the SNR of the known recovered companions and with the planet detection
344 sensitivity at multiple separations from the star. We perform our KLIP subtractions in

345 accordance with the optimal set of KLIP hyperparameters described in the respective
346 publications.

347 For TYC 5899, the target with no known companions, we inject planets at various
348 separations and brightnesses to compare the performance of KLIP and FLIP in a
349 controlled setting.

350 **2.2.1 Data preparation**

351 The observations of HIP 65426, 51 Eridani and TYC 5899 and their reference stars
352 were processed with the JWST pipeline and the community-developed tool `spaceKLIP`.
353 Data processing steps (which include ramp-fitting, converting from detector counts to
354 physical flux units and bad pixel cleaning) and observation set up are detailed in the
355 respective papers for those observations^{34,36,37}.

356 The JWST and `spaceKLIP` pipelines include bad pixel flagging and cleaning. How-
357 ever, imperfect bad pixel cleaning can cause more difficulties in our model-based PSF
358 subtraction than in methods like KLIP. This is because hot or dead pixels, or vari-
359 ations in pixel responses, cannot be “canceled-out” by subtracting observations with
360 the same pixel variations. Therefore, bad pixels that persist even after the automated
361 cleaning routines are identified by eye and masked during our optimization process.

362 **2.2.2 Choice of reference stars**

363 Both HIP 65426 and 51 Eridani were observations carried out with dedicated reference
364 stars: a bright star with likely no planets around it, of similar spectral type to the
365 science target, observed right before or after the science observation. For these targets,
366 we use as our calibration star (optimization stage 1, Extended Data Figure 5) the
367 same reference stars used for those observations: HIP 68245 for HIP 65426, and HD
368 30562 for 51 Eridani. See^{34,36} for details.

369 For TYC 5899, the observation was taken as part of a self-referenced program,
370 where only science observations are taken, and a large number of these are put together

371 to be used as a reference library for every other science observation. This strategy
372 increases observing efficiency (hence why many JWST exoplanet discovery surveys
373 are done in this manner), at the cost of lower fidelity reference stars. For an apples-
374 to-apples comparison of FLIP and KLIP, we chose one of the observations in the
375 self-referenced library as our calibration star. We select the HIP 17695 observation
376 as it has the most similar spectral type to TYC 5899 from the targets in the library.
377 See³⁷ for details.

378 **2.2.3 Optimization hyperparameters**

379 Small changes to optimization hyperparameters were done per experiment for optimal
380 results. In particular, the exact number of iterations and the reduced learning rate in
381 stage 2. For HIP 65426, we set the total number of iterations in the learning scheduler
382 to 15000 (peak SNR in our synthetic injections was achieved in 11000 iterations)
383 and the reduced learning rate to 0.5% of its original value. For 51 Eridani, we set
384 the total number of iterations in the learning scheduler to 18000 (peak SNR in our
385 synthetic injections was achieved in 13000 and 16000 iterations for F410M and F430M,
386 respectively) and the reduced learning rate to 0.1% of its original value. For TYC
387 5899, we set the total number of iterations in the learning scheduler to 20000 (peak
388 SNR in our synthetic injections was achieved in 17500 iterations) and the reduced
389 learning rate to 1% of its original value.

390 **2.3 PSF subtraction performance metrics**

391 All KLIP subtractions were carried out with the KLIP parameters specified in
392 the respective publications for optimal performance. In all cases, we obtain PSF
393 subtractions comparable to the published works.

394 The contrast curves for both KLIP and FLIP were calculated by dividing the
395 image into small concentric annuli and computing the standard deviation within each
396 annulus accounting for small sample statistics⁴⁷ and for planet signal loss due to both

algorithm overfitting and coronagraph throughput. We also smooth all datasets with a Gaussian kernel ($\sigma = 1$ px) before computing the standard deviations, since FLIP residuals are mostly dominated by pixel-to-pixel noise. For HIP6546, the contrast curve for our method excludes the central 0.6 arcseconds of the image given that a bright lobe of the planet PSF prevents sampling this region; future work incorporating planet modeling and subtraction will overcome this limitation. See^{34,36,37} for details on computing and calibrating these contrast curves.

For the fundamental sensitivity curve, we use the **ERR** extensions of the JWST data products, which track the combined total uncertainty in each pixel value from read noise and Poisson noise. We generate a random Gaussian sample with mean 0 and σ equal to the total noise level in each pixel. We finally compute the contrast curve on this image in the same way as described above for KLIP and FLIP subtractions, assuming no planet signal loss.

To compute the SNR of each planet detection in Figure 2, we extract the peak flux of the planet by fitting a Gaussian to the core of the planet PSF and divide it by the standard deviation of pixels within an annulus at the separation of the planet and with width of 0.2 arcseconds. To mitigate biases from subtraction-induced PSF deformations and to smooth out pixel-to-pixel noise, we apply a low-pass filter ($\sigma = 1$ px) to all observations before computing the SNR. This process is applied to both KLIP and FLIP subtractions for consistency.

2.3.1 TYC 5899 synthetic injections

To compare the sensitivity of FLIP and KLIP on TYC 5899, we inject synthetic planets with known flux across the image and assess the recovery PSF subtraction. Extended Data Figures 7 and 8 show a range of planet injections, complementing Figure 2. At each separation we inject a planet with flux equal to 80 times the fundamental noise limit.

423 For PSF subtraction, we process all observations in Extended Data Figures 7 and
424 8 identically, using the same learning rates and numbers of iterations for FLIP, and
425 the same numbers of annuli and KL modes for KLIP. While this may not be optimal
426 for exoplanet recovery given the differences in signal loss across the image, we choose
427 to display this for uniformity.

428 We find that FLIP outperforms standard method KLIP in the vast majority of
429 cases, producing higher SNR on a diverse range of possible planets. This illustrates
430 the exoplanet discovery potential of FLIP.

431 2.4 Future improvement details

432 While our experiments show significant improvements relative to standard techniques
433 on a variety of datasets, FLIP has not reached the fundamental noise limit yet and
434 it still causes some planet signal loss. The findings in this work lead to the future
435 avenues for improvement shown below:

436 **Detector model:** Our simplified detector model is sufficient to model the majority
437 of observations taken by JWST NIRCам. However, observations of very bright stars or
438 very deep observations require a more sophisticated treatment of the complex detector
439 effects present in JWST³⁸. This is because pixel-level defects or pipeline-induced arti-
440 facts are significantly above the noise limit and accurate modeling becomes essential.
441 Sophisticated effective detector models or models of the data pre-processing pipeline
442 (following²⁸) are promising to push this technique to the very high contrast regime.

443 **Inferred instrument parameters:** Coronagraphy is an ill-posed and degener-
444 ate problem^{21,22}, and as such there is a wide range of solutions that may produce
445 effectively the same PSF estimate. Furthermore, the nominal STPSF models for JWST
446 NIRCам, which we use to initialize our differentiable model, are known to be imper-
447 fect. To use FLIP to retrieve realistic instrument parameters future calibration work
448 is required. This could include placing strong regularization priors on some of the

449 model parameters (e.g., ⁴⁸), jointly fitting a large number of observations or processing
 450 datasets with multiple stars in the field of view with multiple point source models. Suc-
 451 cessful calibration will enable retrieving the state of the instrument during any science
 452 observations in a physically realistic way, contributing to more accurate understanding
 453 and modeling of JWST.

454 **Computational cost:** For the datasets analyzed in this work, the runtime of
 455 FLIP on an NVIDIA A30 GPU varies from 15 to 35 minutes. This is very efficient for
 456 a model with 10^6 free parameters, but exoplanet imaging routinely requires multiple
 457 runs (e.g., to calibrate total sensitivity as in Figure 3). Addressing systematic model
 458 mismatch (improving our initial conditions) and implementing regularization on some
 459 parameters are two ways to potentially improve the speed of FLIP, perhaps close to
 460 the one minute runtime shown in monochromatic simulated data³⁰.

461 **Scenes with bright astrophysical sources:** For simplicity, the model presented
 462 here only models the stellar PSF of the host star. As seen in Figure 2, however, in
 463 some cases other astrophysical sources, such as bright off-axis stars, close binaries
 464 or debris disks are also present in the scenes. With no other constraints, our model
 465 ends up partially distorting these other objects. Future work on model calibration and
 466 parameter regularization could slightly reduce the flexibility of our model to avoid
 467 distorting other components in the scene. Alternatively, joint modeling of both the
 468 host star and these objects is promising for simultaneously minimizing signal distortion
 469 and characterizing the properties of these objects.

470 **Applicability to ground-based observations:** In a space-based instrument
 471 like JWST, the dominant source of instability for most datasets comes from drift
 472 between the reference star observation and the science observation. For this reason
 473 we obtain good results performing our optimization only once or only a handful of
 474 times per dataset (averaging all or a few subsets of the frames of the observation).

475 However, atmospheric turbulence in ground-based observations evolves on much faster
476 timescales, and so does the stellar PSF.

477 While FLIP can be applied to ground-based observatories in principle, simply for-
478 ward modeling a larger number of wavefront sensing measurements quickly becomes
479 very computationally expensive: ground-based adaptive optics systems measure wave-
480 front errors roughly every millisecond during every science observation, while JWST
481 does it only once roughly every two days. Wavefront sensing data, however, should
482 also significantly help constrain the problem, since they are taken simultaneously with
483 the science observation, unlike JWST.

484 Future work addressing this data volume challenge will be critical to extend this
485 framework to ground-based telescopes. For example, empirical methods have been able
486 to partially manage the high dimensionality of the problem using sparsity regularization
487 (e.g., ⁴⁹). Similar approaches could be used in future iterations of FLIP to reduce the
488 computational load on ground-based datasets.

489 **2.5 Data availability**

490 The JWST NIRCcam data used in this paper are from JWST programs ERS 1386
491 (PI: S. Hinkley), GTO 1412 (PI: M. Perrin) and GTO 1184 (PI: J. Schlieder) and are
492 publicly available from the Mikulski Archive for Space Telescopes (MAST).

493 **3 Extended Data**

494 **Declarations**

495 The authors declare no competing interests.

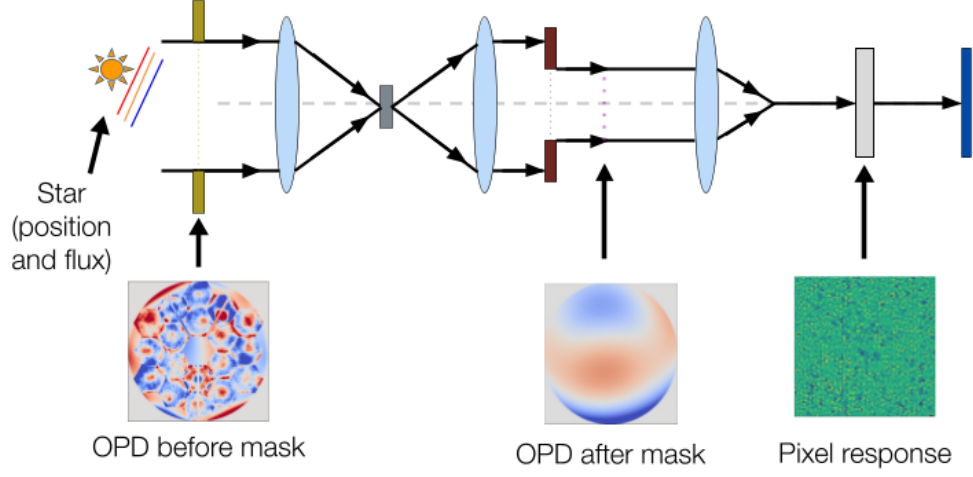
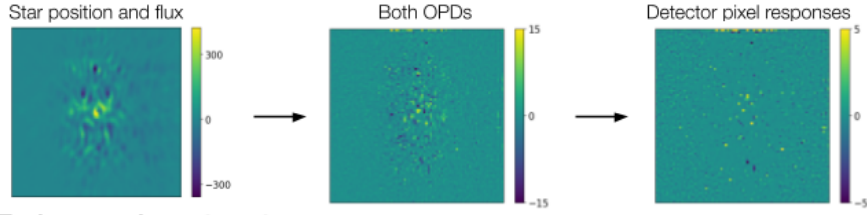


Fig. 4 Model diagram. Schematic of the JWST coronagraph optical model, highlighting the parameters we tune in our model.

1. Calibration on reference star



2. Tuning on science target

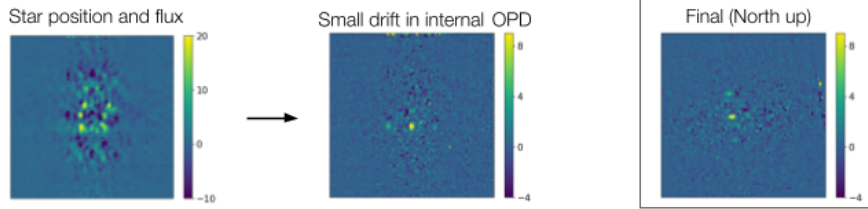


Fig. 5 Optimization framework. Summary of optimization stages, including calibration (stage 1) on a reference star and parameter tuning (stage 2) on the science target of interest. The color scale is adjusted in every step to show the dominant features of the residuals. The example shown is 51 Eridani in the F410M filter, revealing 51 Eridani b in the last panel. For visualization purposes we keep all panels in the detector orientation, except for the last one where we rotate it North up as in Figure 2.

Parameter	Physical interpretation	Dimensions	Learning rate order of magnitude
Star position	Precise location of the star (x,y) behind the coronagraph occulting mask	(2)	10^{-8}
OPD before mask	Deformations or tilts (phase errors) in the telescope mirrors relative to the closest-in-time OPD measurement	(1024,1024)	10^{-9}
OPD after mask	Unknown phase aberrations internal to the JWST NIRCam coronagraph being used	(1024,1024)	10^{-9}
Detector response	Unknown errors in detector calibrations	(80,80)	10^{-5}

Table 1 Model parameters. Summary of free parameters used in our model, their physical meaning and characteristic learning rates during our optimization. The dimensionality of the detector response parameters depends on the field of view being optimized over (in our case, 80×80 pixels in all cases). The flux of the star is scaled automatically to match the total flux across the image during the first sub-stage of stages 1 and 2. As mentioned in the text, while other physical mechanisms can influence the stellar PSF, we find that these four groups of parameters are sufficient to model the PSF with very high accuracy. For this reason, in this work the inferred parameters do not necessarily reflect the true state of the instrument, rather they produce an effective model that is flexible enough to reproduce a variety of systematics. See Section 2.4.

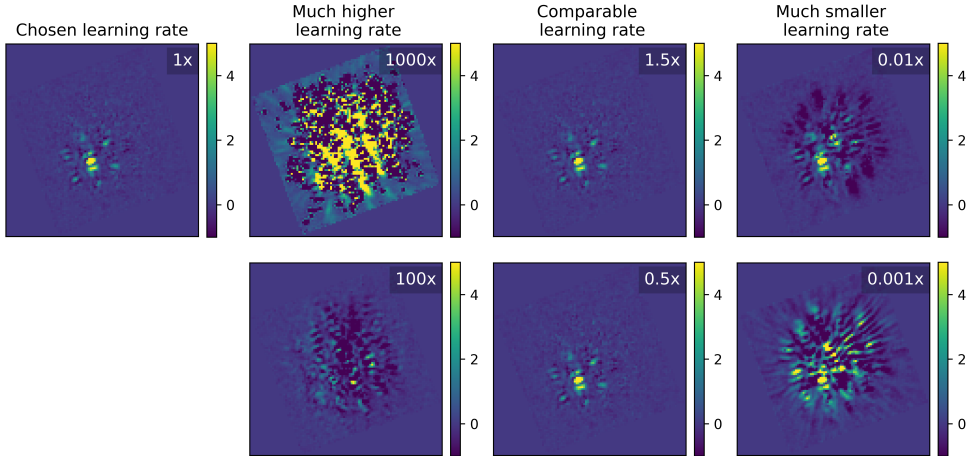


Fig. 6 Sensitivity to chosen learning rates. Example FLIP runs using different learning rates while keeping number of iterations identical. We find that as long as our learning rates are within the correct order of magnitude, the PSF subtraction result is not strongly dependent on the specific choice of the learning rate. The appropriate order of magnitude for the learning rates is obvious after a coarse search. In each panel, all learning rates in the model are scaled by the factor indicated inside the panel.

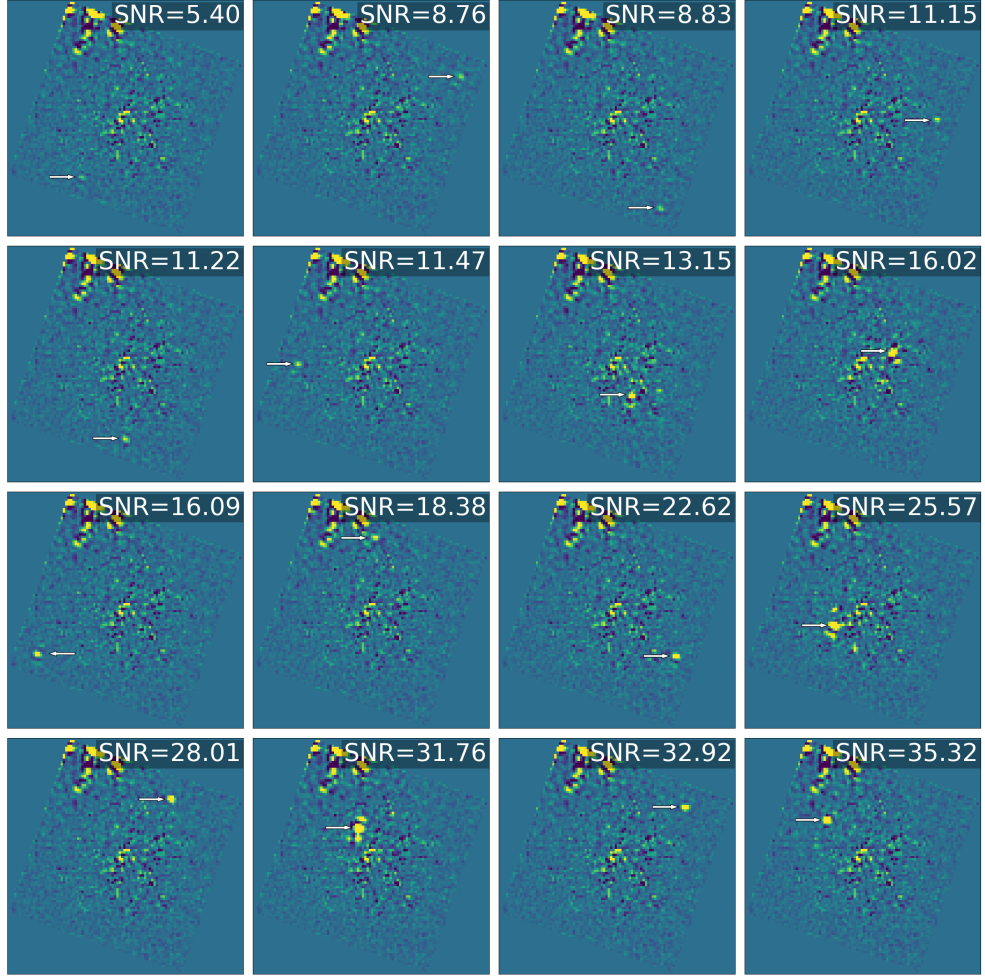


Fig. 7 Suite of injection and recovery tests on TYC 5899 using our method FLIP for PSF subtraction. The location of the synthetic planet is annotated with an arrow. The color scale is identical in all panels and it is the same as for the TYC 5899 observation in Figure 2. The nearby off-axis bright star is also visible.

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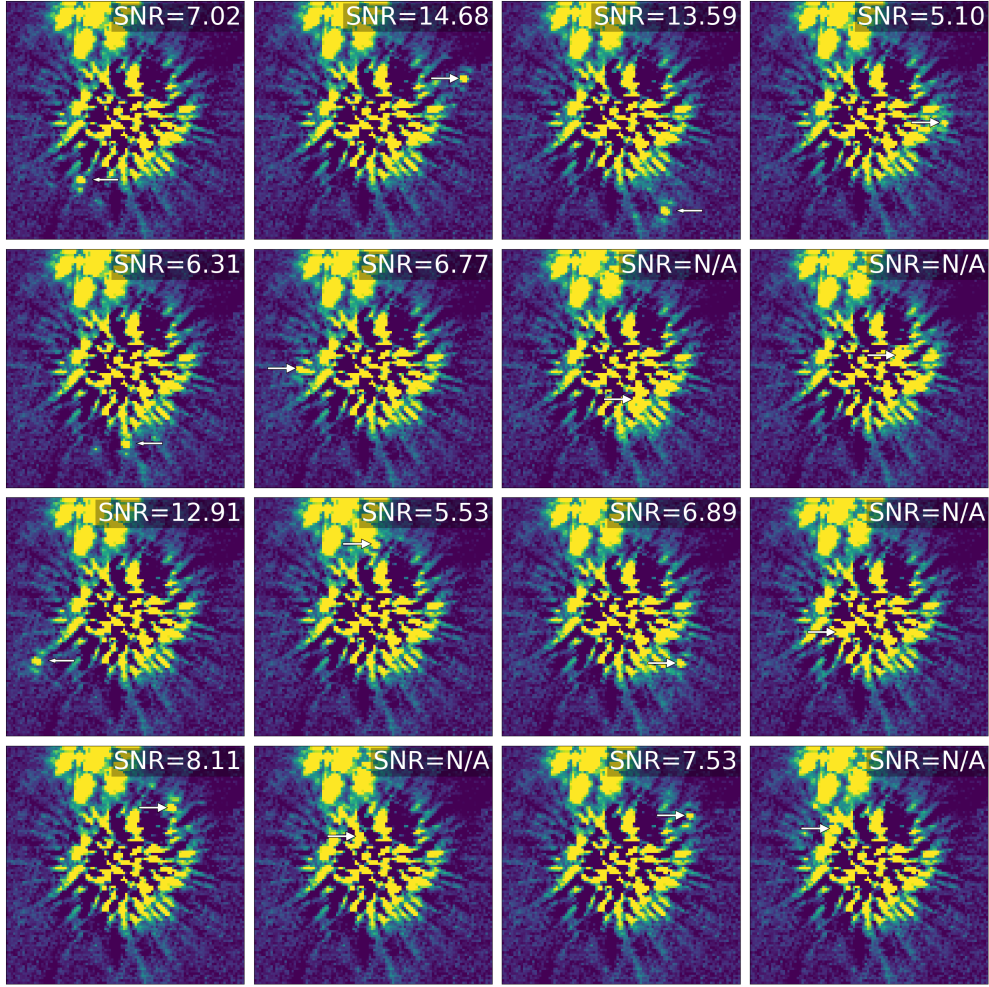


Fig. 8 Same as Figure 7, but using KLIP for PSF subtraction. All injections that resulted in SNR lower than 5 are labeled as SNR=N/A.

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